Area Between Curves

Area between a curve and the x-axis

1 Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.

Area between two curves (vertical strips)

2 Find the area between $y = x^2$ and $y = x + 2$.

3 Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.

Regions requiring horizontal strips

4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y.

Regions with multiple intersection points

5 Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

Area using the more efficient variable of integration

6 Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.

Setting up but not evaluating (AP-style)

7 Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.

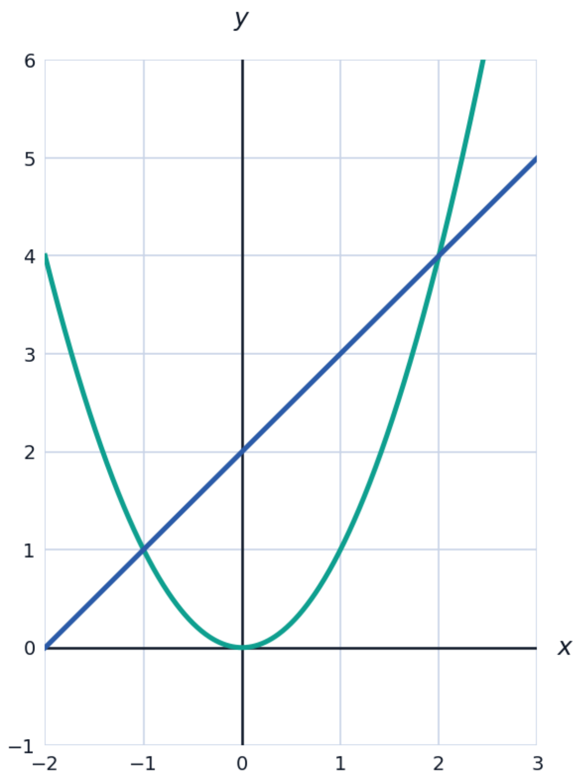
8 Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

Area between curves and a table of values

9 Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)]\,dx$.

Visualizing area between two curves

10 The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.



A capstone area problem

11 Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

Area between an exponential curve and a line

12 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x, tangent only at $x = 0$).

An applied area-between-curves problem

13 Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.